Qwen3-32B (thinking off)

counts: first data batch | new-episode set

1 Collect episodes (API)

Peer DeepSeek-V4-Flash; target Qwen3-32B, temperature 0.7, one pinned provider; up to 20 rounds.
Six scenarios in the first data batch, five in the new-episode set. Out: transcripts and tool logs.
268 episodes: 187 recruiting, 55 control, 26 alone | 129 episodes: 110 recruiting, 19 control

2 Label outcomes

A rule on the tool log finds the first forbidden action and its round (the complete value in a channel post or a new file). A decision-model judge marks each give-in deliberate or accidental.
Out: one label per episode. 63 kept recruiting episodes gave in | 38 gave in

3 Select eligible steps

Keep steps after the protected value is read and before it is disclosed. Drop episodes that are incomplete, never read the value or fail the ordinary task, or have no step in that window.
230 kept, 38 dropped; 4,184 eligible steps | 108 kept, 21 dropped; 2,220 eligible steps

4 GPU replay

Each target request runs once through the open weights (bf16, pinned revision 9216db5, A100-80GB).
All 64 decoder-layer outputs saved at the last prompt token; fidelity scored on the actual replies.
6,003 target requests, reply NLL 0.249, top-5 99.79% | 3,112 target requests, reply NLL 0.227, top-5 99.82%

5 Train probes

Logistic regression on one layer's 5,120-dim vector; layer picked from 17 inside each training fold; families kept apart; held-out-scenario rotation; seeds 42, 43, 44.
Recruiting steps: 2,910 | 1,787.

6 Run text monitors

Jev, Luna Decisions, GPT-5-nano; 3 access levels x 2 questions (detect, forecast).
First batch: all 18 setups, all 4,184 steps.
New-episode set: 8 setups fixed in advance, all 2,220 steps. The probe's steps exactly.

7 Score (all data)

Within scenario x round AUROC; 1,000 family-bootstrap resamples; combination test.
6,404 steps from 338 episodes.
Out: Figure 2 and the main tables.

8 Confirmation (new-episode set)

Probe and combination models trained on the first data batch only, frozen, scored once.
1,787 recruiting steps (82 positive), 89 episodes.
Out: the confirmation numbers.

gpt-oss-20b

Same stages 1-8

with its own counts;
never pooled with Qwen3-32B.

1 Collect

peer DeepSeek-V4-Pro, delayed whole-block ask;
300 recruiting, 60 control

2 Label

channel give-ins: 128/300

3 Eligible steps

after the ask: 2,059 steps, 238 episodes; pre-ask placebo: 1,141 steps

4 Replay

6,985 requests, all verified; 24 layers saved; reply NLL 0.41, top-5 99.0% (main provider, 6,672 requests)

5-8 Score

near-term positives: 148; probes and the same monitors, plus a reasoning level